$(x^T w) =$ $\neq 1$

supervised learning $\frac{(X^T w) w}{\|w\|^2}$

$\|(X^T w) w\| = 84.94$

$x_1 \to y_1$, $x_2 \to y_2$, $\vdots$, $x_n \to y_n$

→ distance
→ size
→ floor

$\frac{(7)(1)}{2}$ $\begin{pmatrix}3\\4\end{pmatrix} - \begin{pmatrix}3.5\\3.5\end{pmatrix}$

$y_1 \ldots y_n \in \mathbb{R}$

reg..

$\boxed{x}$ → 1 $\frac{-2.5}{0}$ -0.5 weak → 0

Regression

→ x_i (100, 95, 80, ...) → (sc, co, run) $y \to (\;)$

over fitting → very low training error

$y_1, y_2 \ldots y_n \in \{0,1\}$

$x_1, x_2, \ldots, x_n$ → 2

$w^T x > 0$ → 1, < -1 → 0

$y_1 \in \{1, \ldots, k\}$ → k-classes

x_i 36 33 = 3 9 x_i 30 33 = 3 9

Regression

$y \ni$ label → dependent on Dataset

$y_i = f(x_i)$

↳ learn this function from our Dataset

measure of goodness

$\sum_{i=1}^{n} (f(x_i) - y_i)^2$ → squared error

(minimum)

$f(x_i) \to$ $x \in \mathbb{R}^d$

training error → y_i

$f(x_i) = y_i$ ↳ is there any

(x_{test}) test error

def $f(x_i)$
if $f(x_i) == (1, 2, 3, 4)$:
return y_i

$k \ni (x_1, \ldots, x_n)$ $y = w_1 x_1 + w_2 x_2 + \ldots + w_d x_d + c$

$w^T x + c$

$w = \begin{pmatrix} w_0 \\ w_1 \\ \vdots \\ w_d \end{pmatrix}$ $x = \begin{pmatrix} x_1 \\ \vdots \\ x_d \end{pmatrix}$

$y = w^T x + c$

$y = f(x)$
$= w^T x$

$y = 10x$, $w_0 = 30$

$y = 3x + 5$ $x \in$ x-axis

$w = 3$, $x = 3$ $x = \begin{pmatrix} 1 \\ x \end{pmatrix}$

$x = 3$ $x = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$ $w = \begin{pmatrix} w_0 \\ 3 \end{pmatrix}$

$\begin{pmatrix} 1 \\ x_2 \end{pmatrix} \begin{pmatrix} 1 \\ x_3 \end{pmatrix} \ldots \begin{pmatrix} 1 \\ x_n \end{pmatrix}$

$w^T x$

$w_0(1) + w_1(x_1)$

$c = m x$

$\sum_{i=1}^{n} (f(x) - y)^2 \Rightarrow \sum_{i=1}^{n} (w^T x_i - y_i)^2 \Rightarrow$ find w which minimizes this

↳ squared error

$(w_1, w_2, w_3, \ldots, w_n)$

Dataset $= (x_1 \ldots x_n)$ $x_i \in \mathbb{R}^d$
label $= (y_1 \ldots y_n) \to \in \mathbb{R}$

find $w \to (w_1 \ldots w_d)$

have we learnt w s.t squared-error is min

linear Regression

$w \to$

→ our assumption of a linear relationship b/w y & x is true

$\mathbb{R}^d$ $x \to (x_1 \ldots x_d, y_i)$
y

lie on a

every d-data point satisfies hyperplane $\mathbb{R}^{d+1}$

$w^T x_i = y$

$X^T = \begin{bmatrix} - x_1 - \\ - x_2 - \\ \vdots \\ - x_n - \end{bmatrix}$

$w^T x_i - y_i$

$\sum_{i=1}^{n} (w^T x_i - y_i)^2$

$\|X^T w - y\|^2$

$y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$

$w^T x = x^T w$

$x_i^T w - y$

$X^T w$

$\begin{pmatrix} x_1^T w - y_1 \\ x_2^T w - y_2 \\ \vdots \\ x_n^T w - y_n \end{pmatrix}^2 + 2$

$d \times d$

$w^T x = x^T w$

$\|X^T w - y\|^2$

$(X^T w - y)^T (X^T w - y)$

$(w^T x)(X^T w - y) - (y^T)(X^T w - y)$

$w^T x x^T w - w^T x y - y^T x^T w - y^T y$

$w^T x y$

$d \times d$

$w^T x = x^T w$

$(w^T x)(x^T w - y) = (y^T)(x^T w - y)$

$w^T x x^T w - w^T x y - y^T x^T w - y^T y$

$w^T x x^T w$

$2 x x^T w$

$-2 w^T x y$

$-2 x y$

$(w_1 \dots w_d)$

$w_i^2 \; x_i^2$

$w^T x x^T w - 2 w^T x y - y^T y$

no constraint on w

$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \Rightarrow$

$\nabla g = (__, __, \dots)$

$\nabla \Rightarrow 2 x x^T w - 2 x y$

$\cancel{2} x x^T w - \cancel{2} x y = 0$

$x x^T w = x y$

$(x x^T)^+ (x x^T) w = (x x^T)^+ x y$

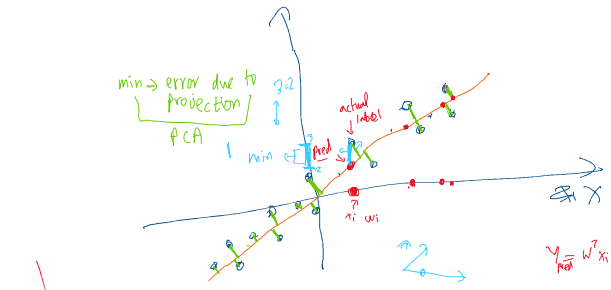
$C = \frac{1}{n} x x^T$

scaled

depends on labels

$w = (x x^T)^+ x y$

$\boxed{w = (x x^T)^+ x y}$



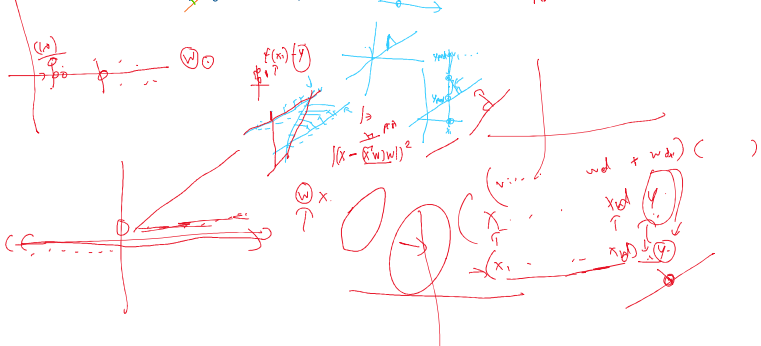
$x_1 \dots x_n \Rightarrow \mathbb{R}$

$y_1 \dots y_n \Rightarrow \mathbb{R}$

$v_1 \quad v_n \in \mathbb{R}^2$

$\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} \quad \begin{pmatrix} x_n \\ y_n \end{pmatrix}$

$y_{pred} = w^T x_i$



$f_1 \; f_2 \dots f_d$

$x_1 \quad y$

$x_2 \quad y_2$

$x_n \quad y_n$

$f_1 \; f_2 \dots f_n$

$x_1 \; x_2 \; x_n$

$y_1 \; y_2 \; y_n$

y

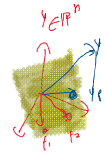
$f_1 \; f_2 \dots f_n$

$f_1 \; f_2$

$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} \in \mathbb{R}^n$

$\mathbb{R}^3 \quad \mathbb{R}^3$

$y \in \mathbb{R}^n$



span $(f_1, f_2) \Rightarrow$ 2-d subspace in $\mathbb{R}^3$

$y_p = \alpha_1 f_1 + \alpha_2 f_2$

$y_p = \begin{bmatrix} f_1 & f_2 \end{bmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}$

projection of y

linear combination

$x_1 \rightarrow$

$x_2 \rightarrow$

$x_3 \rightarrow$

* Projection of y is a linear combination of feature

$$* \; y_p = \begin{bmatrix} f_1 & f_2 \end{bmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} \qquad \begin{array}{c} x_1 \to \\ x_2 \to \\ x_3 \to \end{array} \begin{bmatrix} f_1 & f_2 \\ | & | \end{bmatrix}$$

$$y_p = X^T \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}$$

$\underline{(x - x_p)} + x_p = x$

$$\begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} \qquad y_p = X^T \underset{w}{\alpha} \qquad y_p - X^T\alpha$$

$$(y_p - X^T\alpha)^T (X^T\alpha) = 0$$

$$\alpha = w^* \leftarrow \qquad \alpha = \underline{w^*}$$

$$= (XX^T)^{+} Xy$$

$$(y_p - \underbrace{X^T (XX^T)^{+} Xy}_{(XX^T)^{+}})^T (X^T (XX^T)^{+} Xy) = 0$$

$$y^T X^T \underbrace{(XX^T)^{+}}_{(XX^T)^{+}})^T X$$

$$y^T X^T (XX^T)^{+} X$$

Take away

Projection of y onto span$(f_1 \cdots f_d)$ is $(X^T w)$

$$\boxed{X^T w^*} \to \text{projection of } y \text{ onto span}(f_1, \cdots f_d)$$

$$\|y - \underbrace{(X^T w)}_{\text{same data points}}\|^2 = 0$$

$v_1 x_1 + v_2 x_2 + \cdots v_d x_d$